

TESTER SUPREME

0.0 PIPELINE LAB

Blender-to-s&box Asset Pipeline and Pre-Production Plan

Project foundation before gameplay development

Plan item	Decision
Art direction	Goofy, stylized mid-poly; exaggerated but grounded
Target platform	PC only
Pipeline	Blender → FBX → ModelDoc → s&box scene/prefab
Primary goal	A reliable, versioned modular-model workflow at correct scale, built to hold up as the framework grows past this lab
First gameplay gate	Player movement and one basic interaction sequence

Version 2.0 | July 2026

Internal working guide — supersedes Version 1.0

What changed in this revision

Version 2.0 folds in a scaling and robustness pass: version control for binary assets, engine/tool version pinning, a collision-authoring convention, texel density and lightmap UV standards, poly budget targets, template versioning, an automation roadmap, and a risk register. See Appendix E for the full revision history and why each change was made.

01 Purpose and Success Definition

This plan establishes the production pipeline before Tester Supreme gameplay begins.

Tester Supreme 0.0 is a controlled pre-production phase. Its purpose is to prove that assets can be modeled in Blender using inches, exported consistently, imported into s&b at the correct scale and orientation, converted into reusable prefabs, and assembled into modular rooms, and used safely by the built-in player controller.

The governing rule

Gameplay production does not begin until a new model can travel from Blender to a usable s&b prefab without repeated manual correction.

WHAT THE PIPELINE MUST PROVE

- The official Citizen human reference can be used in Blender as a reliable visual scale guide.
- A verified FBX export preset produces correct scale, orientation, pivots, normals and material slots.
- Modular floors, walls, corners, doorways and ceilings connect without visible gaps or collision seams.
- Shared s&b materials can be assigned and survive reimporting.
- Prefab instances update when the source model is revised.
- The built-in player controller can walk, jump, fall, use stairs, cross seams and interact with the test kit.
- The workflow stays reproducible and low-friction as the number of modular assets grows well past the size of this test kit — see Section 16.

OUTSIDE THIS PHASE

- Weapons, enemies, progression, scoring and narrative content.
- Final visual polish, unique wear, complex decals or production-quality texture variation.
- Multiplayer, networking, monetization or public release preparation.
- Large-scale architecture production before scale and snapping have been validated.
- Performance optimization, LODs and draw-distance culling.

02 Internal Version Roadmap

Version	Milestone	Definition
0.0.1	Foundation	Folders, empty s&box project, Blender template, Citizen reference, scale and FBX calibration, version control and environment baseline.
0.0.2	Modular Test Kit	First architecture, movement, material and interactive models imported and configured.
0.0.3	Pipeline Lab	Modular scenario assembled; player movement, jump, fall, stairs, ramps and seams tested.
0.0.4	Interaction and Reimport	Button, warning light, sliding door, physics crate and source-model update tests.
0.0.5	Production Kit Start	First stylized Calibration Wing modular family built from the approved template.
0.1	Public Calibration Wing	A complete miniature game with three experiments, progression and a clear ending.

Public-versus-internal boundary

Versions 0.0.1 through 0.0.5 are development milestones. Version 0.1 is the first public-facing early-development build.

Automation introduced in Section 16 begins informally at 0.0.2 (batch export) and is expected to mature through 0.0.5 as the number of tracked assets grows beyond what manual checklists can reliably cover.

03 Master Folder Structure

Keep runtime assets, editable source files and project documentation separate.

```
TesterSupreme/  
├─ Game/  
│   ├── Assets/  
│   │   ├── models/  
│   │   │   ├── calibration/  
│   │   │   ├── architecture/  
│   │   │   ├── props/  
│   │   │   ├── interactive/  
│   │   │   ├── characters/  
│   │   │   └─ weapons/  
│   │   ├── materials/  
│   │   ├── textures/  
│   │   ├── prefabs/  
│   │   ├── scenes/  
│   │   ├── sounds/  
│   │   ├── particles/  
│   │   └─ ui/  
│   ├── Code/  
│   │   ├── Core/  
│   │   ├── Interaction/  
│   │   ├── Player/  
│   │   ├── Testing/  
│   │   └─ Debug/  
│   └─ Libraries/  
├─ Production/  
│   ├── Blender/  
│   │   ├── Templates/  
│   │   ├── Calibration/  
│   │   ├── Architecture/  
│   │   ├── Props/  
│   │   ├── Interactive/  
│   │   ├── Characters/  
│   │   └─ Weapons/  
│   ├── References/  
│   │   ├── SboxCitizen/  
│   │   ├── ArtDirection/  
│   │   └─ Measurements/  
│   ├── Textures/  
│   │   ├── Source/  
│   │   ├── Bakes/  
│   │   └─ Export/
```

```
|   └─ ExportArchive/
|       └─ Models/
|           └─ Textures/
└─ Documentation/
└─ ReferenceScreenshots/
```

Placeholder folders

Characters/, weapons/, sounds/ and **ui/** exist in the scaffold now but have no process defined until their owning phase begins (0.1 and later). Do not populate them with content or ad-hoc conventions during the Pipeline Lab — an empty, reserved folder is intentional, not an oversight.

INITIAL S&BOX SCENES

```
Game/Assets/scenes/development/pipeline_lab.scene
Game/Assets/scenes/tests/scale_test.scene
Game/Assets/scenes/tests/material_test.scene
Game/Assets/scenes/tests/interaction_test.scene
```

REQUIRED DOCUMENTATION FILES

File	Purpose
PIPELINE.md	High-level production process and approval status.
VERSION_CONTROL.md	Git LFS policy, tracked/ignored paths, branching notes. See Section 04.
BLENDER_TEMPLATE.md	Units, collections, scene reference and template rules.
FBX_EXPORT_SETTINGS.md	Verified exporter values, known exceptions and the recorded Blender/s&box environment baseline.
ASSET_NAMING.md	Models, materials, textures, prefabs and scene naming.
MODULAR_DIMENSIONS.md	Approved module dimensions, pivots and snapping rules.
MATERIAL_LIBRARY.md	Shared materials, texture sets, texel density and intended uses.
IMPORT_CHECKLIST.md	Per-asset ModelDoc, collision and prefab checklist.
TEST_LOG.md	Failures, observations, fixes and retest results.

04 Version Control and Environment Baseline

Binary art assets and a fast-moving engine both need explicit rules, or the pipeline quietly drifts.

SOURCE CONTROL FOR BINARY ASSETS

Track .blend, .fbx and image/audio source files with Git LFS rather than plain git. Plain git stores every binary revision in full inside history, which gets slow and heavy fast once the modular kit grows past a handful of assets — exactly the growth this project is planning for.

```
*.blend filter=lfs diff=lfs merge=lfs -text
*.fbx   filter=lfs diff=lfs merge=lfs -text
*.png   filter=lfs diff=lfs merge=lfs -text
*.psd   filter=lfs diff=lfs merge=lfs -text
*.wav   filter=lfs diff=lfs merge=lfs -text
```

- ☐ Git LFS installed and initialized before the first binary asset is committed.
- ☐ .gitattributes (above) committed before the Foundation milestone closes.
- ☐ Compiled or generated s&box output (bin/, obj/, .generated/, cache/) stays out of version control.

ENVIRONMENT PINNING

s&box is under active development and Blender's FBX exporter behavior varies by version. A calibration that passes today can silently fail after either tool updates. Record the environment the calibration was locked against, and re-check it whenever either tool changes.

Field	Recorded value
Blender version	
s&box build / branch	
Recorded date	
Locked by	

Why this matters

A calibration cube that measured correctly last month is not proof it still measures correctly today. Treat any Blender or s&box update as a reason to recheck, not just a reason to keep working.

RECALIBRATION TRIGGER

- ☐ Blender updated since the last recorded environment.
- ☐ s&box updated since the last recorded environment.
- ☐ Re-export and re-measure cal_cube_12in against the current environment.
- ☐ Record the result in TEST_LOG.md even when nothing changed.

05 Create the Empty s&box Project

1. Create a new Game project using the cleanest available empty-project option.
2. Save the first scene as Assets/scenes/development/pipeline_lab.scene.
3. Add basic neutral lighting and a temporary floor.
4. Add the standard playable character/controller from the current sample setup.
5. Confirm that the player can look, walk, jump and fall before importing custom assets.
6. Install and initialize Git LFS, then commit the .gitattributes from Section 04 before adding any binary reference assets.
7. Initialize version control and commit the untouched project baseline.
8. Record the current Blender and s&box versions in FBX_EXPORT_SETTINGS.md as the initial environment baseline.

Use samples as references, not as the project foundation

Copy only the minimum controller setup required for testing. Avoid inheriting a large sample game that obscures which systems belong to Tester Supreme.

06 Citizen Human Reference

Use the official male human reference as the Blender equivalent of the Roblox R15 scale mannequin.

SOURCE AND DESTINATION

Source:

sbox/addons/citizen/Assets/models/citizen_human/citizen_human_male_REF.fbx

Project copy:

TesterSupreme/Production/References/SboxCitizen/citizen_human_male_REF.fbx

- ☐ Copy the reference file; do not move or edit the original installed file.
- ☐ Keep the copied master unchanged and use it only as a visual modeling reference.
- ☐ Never include the Citizen reference in architecture or prop FBX exports.
- ☐ Do not parent modular assets to the Citizen hierarchy.
- ☐ Validate Blender scale against the actual playable s&box controller before locking measurements.

07 Blender Modeling Template

TEMPLATE FILE

Production/Blender/Templates/TS_SBOX_MODELING_TEMPLATE.blend

UNIT SETTINGS

Setting	Initial value
Unit System	Imperial
Length	Inches
Unit Scale	1.000 initially; lock only after calibration
Modeling input	Enter dimensions explicitly in inches
Grid snap increment	8 in — matches wall thickness and the module family in Appendix B

TEMPLATE COLLECTIONS

```
TS_SBOX_MODELING_TEMPLATE.blend
├─ _REFERENCE
│   ├── Citizen_Male
│   ├── Scale_Rulers
│   ├── Axis_Markers
│   └─ Ground_Reference
├─ _WORK
│   ├── Architecture
│   ├── Props
│   └─ Interactive
├─ _COLLISION_REFERENCE
├─ _EXPORT
└─ _TEMP
```

Collection	Use
_REFERENCE	Citizen, rulers, orientation markers and ground reference. Never exported.
_WORK	Current editable production geometry.
_COLLISION_REFERENCE	Optional simplified shapes used while planning collision.
_EXPORT	Only finished objects intended for the current FBX export.

_TEMP

Boolean cutters, backup geometry and experiments. Never exported.

08 Import and Verify the Citizen

1. Import citizen_human_male_REF.fbx into the Blender template.
2. Move all imported objects into _REFERENCE/Citizen_Male.
3. Preserve the imported hierarchy and transforms during the first inspection.
4. Place the visible feet at the $Z = 0$ ground reference.
5. Add a separate 72-inch marker and compare it with the imported body.
6. Record the imported height, object scale and importer settings in BLENDER_TEMPLATE.md.
7. Disable the Citizen collection from rendering and prevent accidental selection.

Calibration warning

The reference FBX may contain unit metadata that Blender interprets differently from new inch-authored assets. Do not force a conversion value until both the reference and an exported calibration cube have been checked in s&bbox.

PERMANENT TEMPLATE REFERENCES

- 12-inch cube
- 24-inch ruler
- 48-inch ruler
- 72-inch human-height marker
- 96-inch height marker
- FRONT arrow
- UP arrow
- Ground plane at $Z = 0$

09 Pivot, Origin and Transform Standards

Asset type	Origin / pivot
Floor tile	Bottom rear-left corner
Wall panel	Bottom rear-left corner at the back face
Ceiling panel	Consistent corner aligned with the floor convention
Freestanding prop	Bottom center
Crate	Bottom center
Door frame	Bottom rear-left corner
Swing door	Exact hinge axis
Sliding door	Center or movement origin used by the animation
Button	Center of pressing motion
Terminal	Bottom center
Rotating machine	Exact mechanical rotation axis

NORMAL EXPORT-READY TRANSFORMS

```
Location: 0, 0, 0
Rotation: 0, 0, 0
Scale:    1, 1, 1
```

Fix bad pivots in Blender

Do not compensate for a wrong origin by repeatedly offsetting every scene instance in s&box. The source asset is the correct place to fix it.

10 Scale and FBX Calibration Set

CALIBRATION BLENDER FILE

Production/Blender/Calibration/TS_EXPORT_CALIBRATION.blend

Asset	Dimensions / form	Test purpose
cal_cube_12in	12 × 12 × 12 inches	Basic dimensional accuracy
cal_floor_96in	96 × 96 × 4 inches	Grid alignment and modular snapping
cal_wall_96x120	96 × 120 × 8 inches	Human scale, doorway and vertical proportion
cal_axis_marker	Asymmetrical labeled object	Forward, up, mirroring and rotation
cal_material_block	Four visible surface zones	Painted, rough, metallic and emissive response

Forward / Up is a hypothesis to test, not an assumption

Do not assume Blender's default Forward/Up mapping matches what s&box's ModelDoc importer expects. Try candidate combinations against the asymmetrical axis marker in Bay A before locking anything, and record the exact combination that measured correctly — never leave it on "auto" once approved. This is one of the most common silent-failure points in a new engine pipeline.

INITIAL BINARY FBX CANDIDATE SETTINGS

Setting	Starting value
Limit To	Selected Objects
Object Types	Mesh
Scale	1.000
Apply Unit	Enabled initially; validate
Apply Transform	Test before locking
Bake Animation	Disabled for static kit
Add Leaf Bones	Disabled
Path Mode	Auto or Strip
Forward / Up	Resolve empirically against the axis marker; record the result

CALIBRATION EXPORT PROCEDURE

1. Select one calibration object and place it in _EXPORT.
2. Apply transforms only after the origin is correct.
3. Inspect face orientation and remove accidental internal geometry.
4. Export one binary FBX per calibration object.
5. Import through ModelDoc and place the model at scene scale 1,1,1.
6. Measure and compare against the s&box grid and playable character.
7. Change the Blender export preset rather than scaling each placed asset.
8. Repeat until re-exporting produces the same size and placement every time.
9. Record the current Blender/s&box versions in FBX_EXPORT_SETTINGS.md alongside the locked preset.

11 Calibration Approval Gate

- ☐ The 12-inch cube measures correctly in s&box.
- ☐ The 96-inch floor aligns with the selected engine grid.
- ☐ The 120-inch wall feels correct beside the actual playable character.
- ☐ The asymmetrical axis marker faces the expected direction and is not mirrored.
- ☐ Every placed model remains at scale 1,1,1.
- ☐ Two floor modules meet without gaps or overlap.
- ☐ Two wall modules meet without gaps or overlap.
- ☐ Re-exporting the source does not move existing instances.
- ☐ Blender and s&box versions are recorded alongside the locked preset.
- ☐ The resolved Forward/Up combination is recorded explicitly, not left as "auto".

Lock only after passing

Save the verified export preset, update the documentation, and archive TS_SBOX_MODELING_TEMPLATE_v1.blend only after every calibration item passes.

TEMPLATE VERSIONING AFTER V1 IS LOCKED

1. Do not edit TS_SBOX_MODELING_TEMPLATE_v1.blend in place.
2. Duplicate it to a new versioned file (_v2) and make changes there.
3. Re-run this entire approval checklist against the new version.
4. Re-export and re-verify at least one already-approved test-kit asset against the new template before adopting it project-wide.
5. Record the version change and the reason for it in TEST_LOG.md.

A newer template is not automatically better

Nothing built against v1 should be assumed compatible with v2 until Bay F's reimport test (Section 24) has been repeated against it.

12 First Modular Test Kit

Reuse the approved template to create a deliberately small, goofy mid-poly laboratory kit.

WORKING FILE

Production/Blender/Calibration/TS_MODULAR_TEST_KIT.blend

Architecture asset	Starting dimensions	Purpose
Floor tile	96 × 96 × 4 in	Primary module
Half floor	48 × 96 × 4 in	Layout flexibility
Solid wall	96 × 120 × 8 in	Primary wall
Half wall	48 × 120 × 8 in	Short span
Doorway wall	96 × 120 × 8 in	Player clearance
Window wall	96 × 120 × 8 in	Visibility and trim
Ceiling tile	96 × 96 × 4 in	Room closure
Inside corner	Fits 8-inch wall family	Interior turns
Outside corner	Fits 8-inch wall family	Exterior turns
Support column	16 × 16 × 120 in	Structural rhythm

Movement asset	Purpose
12-inch crate	Small step and edge test
24-inch crate	Jump and landing test
36-inch crate	High obstruction and climb-limit test
Ramp	Slope handling
Short stair set	Step height and collision
Narrow doorway	Controller width and camera clearance
Low beam	Crouch/camera-height test if supported
Drop platform	Falling and landing behavior

Interactive asset	Purpose
Test button	Basic press/use input
Sliding door	Moving collision and safe passage
Test terminal	Interaction range and prompt placement
Physics crate	Player-to-object physics
Warning light	Immediate state feedback

STYLE RULES FOR THE TEST KIT

- Use thick bevels, overbuilt frames and large readable silhouettes.
- Exaggerate handles, bolts, warning lights and machinery without adding tiny decorative noise.
- Keep architecture reusable; put personality into a few replaceable trim and prop pieces.
- Avoid unique wear and damage until the base material pipeline is proven.
- Use more geometry only where it improves silhouette, curvature, interaction or close-up quality.

POLY BUDGET TARGETS (WORKING)

Asset class	Target triangle ceiling
Floor / wall / ceiling module	≤ 200 tri
Trim / corner piece	≤ 150 tri
Freestanding prop (crate, column)	≤ 400 tri
Interactive prop (button, door, terminal)	≤ 600 tri

Provisional until tested

These are starting ceilings, not final rules. If the goofy stylized silhouette needs more geometry to read correctly, revise the number and record why in TEST_LOG.md rather than quietly exceeding it asset by asset.

13 Material, Texture and Texel Density Standards

Material	Initial use
ts_calibration_painted_metal	Primary wall and equipment shell
ts_calibration_dark_metal	Frames, rails and mechanical structure
ts_calibration_rubber	Feet, grips, bumpers and seals
ts_calibration_emissive	Screens, warning lamps and status lights

TEXTURE NAMING EXAMPLE

```
ts_calibration_metal_color.png
ts_calibration_metal_normal.png
ts_calibration_metal_rough.png
ts_calibration_metal_metal.png
ts_calibration_metal_ao.png
ts_calibration_screen_selfillum.png
```

BLENDER MATERIAL NAMES

```
M_CalibrationPaintedMetal
M_CalibrationDarkMetal
M_CalibrationRubber
M_CalibrationScreen
```

Avoid automatic Blender suffixes

Do not leave names such as Material.001 or Metal.002. Period-based suffixes can create confusing or collapsed imported material slots.

MATERIAL EFFICIENCY RULES

- Background modular pieces should normally use one material.
- Interactive props should normally use one or two materials.
- Add more material slots only when they create a visible gameplay or close-up benefit.
- Use shared materials and consistent texel density instead of unique textures for every wall.

TEXEL DENSITY STANDARD

Setting	Value
---------	-------

Target texel density	8 px per inch (128 px per 16 in) — working value
Reference resolution	1024 × 1024 per shared material sheet
Consistency rule	All modular architecture shares one texel density; hero/interactive props may increase it, never decrease it

LIGHTMAP UV (UV2)

Every architecture module intended for static placement gets a second, non-overlapping UV channel reserved for lightmapping, authored at the same time as UV1 — even though lightmapping itself is not validated until later.

- ☐ UV1 (material) present.
- ☐ UV2 (lightmap, non-overlapping, 0–1 space) present on architecture.
- ☐ UV2 intentionally skipped for movable/interactive props is noted, not silently absent.

Why now

Retrofitting a second UV channel onto an already-approved modular kit later means re-touching every approved asset. Reserving it during the Modular Test Kit milestone costs little and avoids that rework.

14 Import into ModelDoc and s&box

1. Export each reusable test-kit asset as its own binary FBX.
2. Place source files in the matching Game/Assets/models category.
3. Create or update the corresponding .vmdl in ModelDoc.
4. Assign the final s&box material rather than relying on Blender viewport materials.
5. Place the model in a test scene at scale 1,1,1.
6. Confirm size, orientation, origin, normals, smoothing and material slots.
7. Record the current Blender/s&box versions if this is the first import since the last recorded baseline (Section 04).
8. Record any required exception in IMPORT_CHECKLIST.md.

PER-MODEL IMPORT CHECKLIST

- ☐ Correct size
- ☐ Correct forward direction
- ☐ Correct up direction
- ☐ No mirroring
- ☐ Correct pivot
- ☐ Correct normals
- ☐ Correct smoothing
- ☐ No missing faces
- ☐ Expected material slots
- ☐ No unexplained scene scaling

15 Functional Prefabs and Collision Authoring

EXAMPLE PREFAB STRUCTURE

```
PF_CalibrationWall
├─ ModelRenderer
└─ BoxCollider

PF_TestCrate
├─ ModelRenderer
├─ BoxCollider
└─ Rigidbody [movable version only]

PF_TestDoor
├─ DoorFrame
│   ├── ModelRenderer
│   └─ BoxCollider
└─ DoorPanel
    ├── ModelRenderer
    └─ BoxCollider
```

Asset	Initial collision approach
Floors	Simple box collider
Walls	Simple box collider
Crates	Box collider; rigid body only for movable variants
Columns	Box collider
Ramps	Simplest collision shape that matches movement needs
Door frame	Several simple colliders if necessary
Decorative bolts and trim	No individual collision

Collision serves gameplay

The player should not collide with every bolt, recess or decorative lip. Simplify collision until it supports predictable movement.

COLLISION AUTHORING CONVENTION (PER SHAPE)

Shape	Authoring method
-------	------------------

Flat floor / wall / ceiling	Native BoxCollider sized to bounds
Column	Native BoxCollider, or CapsuleCollider if the silhouette reads as round
Ramp	A BoxCollider rotated to match the slope angle — not a flat box placed under it
Stairs	Either one rotated BoxCollider approximating the slope, or one BoxCollider per step — pick one convention project-wide and do not mix them
Door frame	Two BoxColliders (frame sides), matching the hinge/frame pivot convention in Section 09
Non-box / organic prop	A convex hull collider generated from a simplified proxy mesh, never the render mesh

Pick one stair convention now

Mixing per-step colliders and slope colliders across different staircases in the same project creates inconsistent player-movement feel that is hard to diagnose later.

PHYSICS STARTING VALUES (MOVABLE VARIANTS)

Property	Starting value
Mass (physics crate)	20 kg — working value
Friction	Default/medium — tune only if sliding is observed

Provisional until Bay D testing (Section 22); record any change and why in TEST_LOG.md.

PREFAB APPROVAL CHECKS

- ☐ Renderable model is correctly assigned.
- ☐ Collision matches the intended walkable or blocking surface, using the convention above.
- ☐ Static and movable variants are clearly separated.
- ☐ Prefab name follows the naming standard.
- ☐ A source update can propagate without replacing every placed instance.

16 Automation and Scaling Strategy

The Pipeline Lab validates the process by hand once. Tester Supreme's long-term goal is a framework that stays reliable as the asset count grows well past the test kit — that requires reducing how much of the process depends on a human repeating it correctly every time.

WHY THIS MATTERS NOW, NOT LATER

- Bay F's reimport test (Section 24) is correct but manual; it does not scale past a handful of assets without becoming a chore that gets skipped under time pressure.
- Skipped regression checks are exactly how "reliable modular workflow" quietly degrades into "worked once."

RECOMMENDED AUTOMATION, INTRODUCED INCREMENTALLY

Milestone	Automation to introduce
0.0.2 — Modular Test Kit	A Blender script/macro that batch-exports every object in _EXPORT to individual binary FBX files using the locked preset, instead of exporting one at a time by hand.
0.0.4 — Interaction and Reimport	A lightweight regression log (script or spreadsheet) tracking which test-kit assets have been reimport-tested since their last source change.
0.0.5 — Production Kit Start	Extend the naming/pivot/collision conventions into a Blender validation script that flags an object before export if its pivot, scale or collection placement doesn't match Sections 09/15.
0.1+	Revisit whether a fuller automated import/validation pass is worth the investment, based on how much manual regression testing actually cost during 0.0.x.

Automation means removing repetitive risk, not tooling for its own sake

If a milestone's manual checklist is still fast and reliable, don't automate it yet. Automate the steps that are easy to get subtly wrong at scale, first.

17 Risk Register

Named risks get mitigated. Unnamed risks get discovered mid-production.

Risk	Impact if unaddressed	Mitigation
s&b's importer behavior changes under active engine development	Silent calibration drift; assets that measure correctly today measure wrong later	Section 04 environment pinning and recalibration trigger
Blender's FBX exporter behavior differs across versions	Same drift, from the authoring side	Pin and record Blender version; re-verify the preset after upgrading
Manual reimport testing gets skipped under time pressure as the kit grows	Prefab/material drift goes unnoticed until it blocks a later milestone	Section 16 automation roadmap; a skipped Bay F pass is a documented exception, not silence
The template changes after v1 lock without re-validation	Assets built against different template versions behave inconsistently	Section 11 template versioning procedure
A one-off asset breaks naming or pivot convention "just this once"	Corner-solution hacks accumulate and make later batch changes unsafe	Section 09's rule (fix the source asset, not the instance) plus the Section 15 prefab checks
Solo-developer blind spots in subjective pass/fail calls	A wrong call goes unchallenged since there is no second reviewer	Route every subjective fail, and any close call, through TEST_LOG.md's Root Cause field

18 Pipeline Lab Scenario Layout

Build one compact scene divided into purpose-built testing bays.

Bay	Purpose
A - Scale Bay	Citizen/player comparison, rulers, cubes, doorway and wall proportions.
B - Modular Construction Bay	Room assembled from floors, walls, corners, window, ceiling and columns.
C - Movement Course	Corridor, stairs, ramp, steps, jump, low beam, drop and recovery area.
D - Interaction Bay	Button, warning light, sliding door, terminal and physics crate.
E - Material Bay	Neutral, dark, colored, reflective and backlit material inspection.
F - Reimport Bay	Repeated prefab instances used to verify source updates and material persistence.

19 Bay A — Scale Test

- Playable character/controller
- 12-inch cube
- 24-inch and 36-inch crates
- 96-inch floor tile
- 120-inch wall
- Doorway module
- Visible height ruler

QUESTIONS TO ANSWER

- ☐ Does the doorway feel intentionally oversized without feeling wrong?
- ☐ Are buttons, terminals and window heights comfortable for the actual camera?
- ☐ Do crates read at the expected size when viewed in first person?
- ☐ Does the Blender Citizen reference agree closely enough with the playable controller?
- ☐ Does the resolved Forward/Up axis marker (Section 10) still face the expected direction here?

20 Bay B — Modular Construction Test

1. Create a room from four 96-inch floor tiles.
2. Add solid walls, a doorway wall, a window wall and corner pieces.
3. Close the room with ceiling tiles.
4. Place support columns at repeated grid positions.
5. Inspect all seams from both sides and from grazing camera angles.
6. Duplicate and rotate modules without changing their scale.

PASS CONDITIONS

- ☐ No visible gaps between expected connections.
- ☐ No overlapping faces that cause flickering.
- ☐ Pivots make placement predictable.
- ☐ Corner solutions do not require per-instance hacks.
- ☐ Floor and ceiling placement follows the same grid logic.

21 Bay C — Movement Course

Order	Element	Test
1	Flat corridor	Basic forward and lateral movement
2	Narrow doorway	Controller width and camera clipping
3	Short stairs	Step handling and snagging
4	Ramp	Slope movement
5	12-inch step	Automatic stepping behavior
6	24-inch crate	Jump and landing
7	Jumpable gap	Air control and collision edges
8	Low overhead beam	Camera/crouch clearance if applicable
9	Drop platform	Fall and landing behavior
10	Recovery path	Return without reloading the scene

MOVEMENT CHECKLIST

- ☐ Walk across floor seams without catching.
- ☐ Move along walls without shaking or snagging.
- ☐ Pass through intended doorways cleanly.
- ☐ Move up and down stairs and ramps predictably (per the single stair convention chosen in Section 15).
- ☐ Jump onto the intended crate height.
- ☐ Land without falling through seams.
- ☐ Avoid becoming trapped in inside corners.
- ☐ Fall out of bounds and recover safely.

22 Bay D — Basic Interaction Test

REQUIRED INTERACTION SEQUENCE

Step	Expected result
1	Player approaches the test button.
2	A clear interaction prompt becomes available.
3	Player presses the use input.
4	Warning light immediately changes state.
5	Sliding door opens.
6	Player passes through without collision problems.
7	Terminal closes or resets the door.
8	Physics crate can be pushed without unstable movement.

INTERACTION CHECKLIST

- ☐ Prompt appears only at a reasonable distance.
- ☐ Object can be activated from expected approach angles.
- ☐ Object cannot be activated through a wall.
- ☐ Only the intended object activates when several are nearby.
- ☐ Visual state change happens immediately.
- ☐ Door does not permanently trap or crush the player.
- ☐ Scenario can be reset without restarting the entire project.
- ☐ Movable crate does not explode, jitter or tunnel through the floor (compare against Section 15's mass/friction starting values).

23 Bay E — Material Inspection

Lighting condition	What to inspect
Neutral bright light	Base color, normal and roughness accuracy
Dark environment	Emissive readability and black-level response
Colored environmental light	Material consistency under stylized lighting
Direct reflection/highlight	Metalness and roughness behavior
Backlighting	Silhouette, transparency and edge readability

MATERIAL QUESTIONS

- ☐ Does painted metal look coated rather than fully metallic?
- ☐ Do rubber parts remain readable without looking glossy?
- ☐ Are normal details visible but not noisy?
- ☐ Are emissive screens bright enough without washing out the panel?
- ☐ Does the material still support the goofy stylized shape language?
- ☐ Does texel density (Section 13) look consistent between adjacent modules?

24 Bay F — Reimport and Update Test

1. Place several instances of the same wall prefab.
2. Return to Blender and modify one obvious visual detail.
3. Re-export the FBX using the verified preset.
4. Refresh or recompile the imported model.
5. Confirm that every prefab instance updates.
6. Confirm that instance position, rotation and scale remain unchanged.
7. Confirm that material assignments remain connected.
8. Confirm that collision still matches the model.
9. Confirm the lightmap UV2 channel (Section 13), if present, survived the reimport.
10. Repeat once with a material-slot-safe change.
11. Log the result in the regression log introduced in Section 16, even when it passes.

Critical production test

A pipeline that works only for first import is not enough. Tester Supreme depends on being able to revise modular assets without rebuilding the scene.

25 Exact Milestone Task Lists

TESTER SUPREME 0.0.1 - FOUNDATION

- ☐ Create the empty s&box Game project.
- ☐ Create the full master folder structure.
- ☐ Install and configure Git LFS; commit .gitattributes (Section 04).
- ☐ Create pipeline_lab.scene and add neutral lighting.
- ☐ Add the standard playable controller and verify movement.
- ☐ Copy citizen_human_male_REF.fbx into the project reference folder.
- ☐ Create TS_SBOX_MODELING_TEMPLATE.blend.
- ☐ Set Blender to Imperial / Inches and set the grid snap increment.
- ☐ Import and organize the Citizen reference.
- ☐ Create the 12-inch calibration cube and asymmetrical axis marker.
- ☐ Export as binary FBX.
- ☐ Import through ModelDoc.
- ☐ Correct the export preset until scale and orientation require no manual fix.
- ☐ Record the initial Blender and s&box environment baseline.
- ☐ Save the verified template and document every setting.

TESTER SUPREME 0.0.2 - MODULAR TEST KIT

- ☐ Duplicate the verified Blender template.
- ☐ Build the architecture, movement and interactive test models.
- ☐ UV unwrap the kit at the consistent texel density (Section 13).
- ☐ Author non-overlapping UV2 lightmap channels on architecture pieces.
- ☐ Create the four calibration materials and texture set.
- ☐ Confirm each asset against its poly budget target (Section 12).
- ☐ Batch-export test-kit assets using the locked preset (script or manual, per Section 16).
- ☐ Create or update the corresponding .vmdl files.
- ☐ Assign final s&box materials.
- ☐ Add simple colliders using the Section 15 authoring convention.
- ☐ Create static and movable variants where needed.
- ☐ Convert configured GameObjects into prefabs.
- ☐ Assemble the first modular room.

TESTER SUPREME 0.0.3 - MOVEMENT LAB

- ☐ Build the Scale Bay and Modular Construction Bay.
- ☐ Build the full movement course.
- ☐ Walk every seam and wall edge.
- ☐ Test stairs, ramps, doorway width and jump heights.
- ☐ Test falling, landing and recovery.
- ☐ Correct dimensions or collision in the source assets.
- ☐ Record approved architecture dimensions.

TESTER SUPREME 0.0.4 - INTERACTION AND REIMPORT

- ☐ Create one pressable button.
- ☐ Connect the button to a warning light and sliding door.
- ☐ Add one terminal reset action.
- ☐ Add one movable physics crate.
- ☐ Test interaction range and obstruction.
- ☐ Test safe door passage and reset behavior.
- ☐ Confirm collision authoring matches the per-shape convention (Section 15).
- ☐ Revise and reimport one Blender model.
- ☐ Confirm prefab updates, materials and collision remain correct.
- ☐ Log the reimport regression result per asset (Section 16).
- ☐ Mark the pipeline approved or record unresolved blockers.

TESTER SUPREME 0.0.5 - PRODUCTION KIT START

- ☐ Duplicate the approved template and conventions; do not modify the locked 0.0.x calibration files.
- ☐ Replace test-kit placeholder shapes with the first stylized Calibration Wing modular family, keeping the same dimensions, pivots and collision conventions.
- ☐ Re-run Bay B, C, D and F against the new family before treating it as approved.
- ☐ Extend the naming standard and material library for the new asset family.
- ☐ Record any convention exceptions the stylized pass required.

26 Final Pipeline Approval Gate

Area	Approval requirement
Blender	Inches display correctly; Citizen is organized; pivots and export collections are consistent.
FBX	One saved binary preset repeatedly produces correct scale and orientation.
ModelDoc	Models import with expected material slots, normals and no unexplained scaling.
Materials	Shared materials are assigned, visually correct, at the agreed texel density, and survive reimporting.
Prefabs	Configured objects update all placed instances without scene rebuilding.
Collision	Simple collision, built with the Section 15 convention, supports predictable walking, jumping and object movement.
Modularity	Floors, walls, ceilings, corners and doorway pieces connect cleanly.
Player testing	Walk, jump, stairs, ramp, fall, landing and recovery all pass.
Interaction	Button, light, sliding door, terminal and physics crate pass.
Version control	Binary assets tracked via Git LFS; .gitattributes committed; no compiled output tracked.
Environment	Blender and s&box versions recorded and current; recalibration trigger checked.
Automation	At minimum, batch export is in use for the test kit; a manual-only export step is a documented limitation, not silent debt.
Documentation	Settings, measurements, exceptions and fixes are recorded.

Ready for production

The pipeline is approved when a new Blender model can be exported, imported, materialized, collided, converted to a prefab and placed in the scene without rediscovering the process — and without the workflow depending on one person remembering every manual step.

DECISION AFTER APPROVAL

Begin Tester Supreme 0.0.5 by replacing the plain test-kit shapes with the first production-ready Calibration Wing family. Keep the same dimensions, pivots, export preset, material conventions and prefab workflow that passed the Pipeline Lab.

A Naming Standard - Initial Draft

Type	Pattern	Example
Blender file	TS_[CATEGORY]_[SET].blend	TS_ARCH_CalibrationWing.blend
Model source	ts_[category]_[asset].fbx	ts_arch_wall_96.fbx
ModelDoc model	ts_[category]_[asset].vmdl	ts_arch_wall_96.vmdl
Prefab	PF_[Category][Asset]	PF_ArchWall96
Prefab variant	PF_[Category][Asset]_[Variant]	PF_ArchWall96_B
Material	ts_[set]_[surface]	ts_calibration_painted_metal
Texture	ts_[set]_[surface]_[map].png	ts_calibration_metal_rough.png
Scene	[purpose]_[name].scene	tests_scale_test.scene

Prefab names are concatenated PascalCase by design

Unlike files, materials and textures, prefab names intentionally drop the underscore between category and asset. This is a deliberate distinction, not an inconsistency — keep it that way project-wide rather than mixing conventions.

RESERVED SUFFIX MEANINGS

Suffix	Meaning
_B, _C, ...	Visual variant with the same footprint and collision — does not require re-approval (Section 11/15).
_v2, _v3, ...	Revision that changes footprint, pivot or collision — requires re-approval via the Section 11 procedure before replacing placed instances.

B Modular Measurement Working Set

Use	Starting value	Notes
Small increment	12 in	Fine prop and movement calibration
Secondary increment	24 in	Crates, steps and prop spacing
Half module	48 in	Partial floors and walls
Primary module	96 in	Main floors, walls and ceilings
Wall height	120 in	Initial lab room height
Wall thickness	8 in	Initial architectural family; matches the Section 07 grid snap increment

Measurements are provisional until tested

These values are production candidates, not permanent rules. The actual player controller and first-person camera have final authority. Poly budget targets live in Section 12; texel density lives in Section 13 — both are kept there rather than duplicated here so each number has one home.

C Test Log Template

Field	Entry
Date	
Build / commit	
Blender / s&box version	
Scene / asset	
Test performed	
Expected result	
Actual result	
Pass / fail	
Root cause	
Fix applied	
Retest result	
Documentation updated	

D First Work Session Checklist

- ☐ Create the TesterSupreme root folder and master subfolders.
- ☐ Create the empty s&box project and pipeline_lab.scene.
- ☐ Confirm the stock player can walk, jump and fall.
- ☐ Install Git LFS and commit .gitattributes before copying any binary reference.
- ☐ Locate and copy citizen_human_male_REF.fbx.
- ☐ Create TS_SBOX_MODELING_TEMPLATE.blend.
- ☐ Set Imperial / Inches, the grid snap increment, and create the reference collections.
- ☐ Import the Citizen reference and place feet at Z = 0.
- ☐ Create the 12-inch cube, 96-inch floor and axis marker.
- ☐ Perform the first binary FBX export.
- ☐ Import the cube through ModelDoc and record the result.
- ☐ Record the initial Blender and s&box versions.

E Revision History

Version	Date	Summary of changes
1.0	July 2026	Initial pre-production plan authored.
2.0	July 2026	Added Git LFS / version-control policy for binary assets (Section 04); Blender and s&box version pinning with a recalibration trigger (Section 04); a Forward/Up axis hypothesis note instead of an unresolved assumption (Section 10); a template versioning procedure for changes after v1 lock (Section 11); poly budget targets (Section 12); texel density and lightmap UV2 standards (Section 13); a per-shape collision authoring convention and physics starting values (Section 15); an automation and scaling roadmap (Section 16); a risk register (Section 17); a previously-missing 0.0.5 milestone task list (Section 25); version-control and environment rows in the Final Approval Gate (Section 26); a prefab variant naming rule (Appendix A); and the Blender/s&box version field in the test log (Appendix C).

END OF PRE-PRODUCTION PLAN